



ANDERSON SERANGOON JUNIOR COLLEGE

2021 JC 2 PRELIMINARY EXAMINATION

H1 CHEMISTRY

8873/01

Paper 1 Multiple Choice

22 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name and class on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Multiple Choice Answer Sheet

Write your name, class and NRIC / FIN number, **including** the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 Preliminary Exam

Exam Details: H1 Chemistry / Paper 1

Date: 22/09/2021

This document consists of **12** printed pages.

- 1 *Use of the Data Booklet is relevant to this question.*

Neon exists as three isotopes, ^{20}Ne , ^{21}Ne and ^{22}Ne .

How many neutrons are present in a neon atom of the most abundant isotope?

- A** 10 **B** 11 **C** 12 **D** 20

Answer: A

From data booklet, the Ar of Ne is 20.2 and proton number is 10.

=> ^{20}Ne is most abundant compared to ^{21}Ne and ^{22}Ne .

Number of neutrons = $20 - 10 = 10$

- 2 Carbon disulfide, CS_2 , is a poisonous, volatile liquid used in the manufacture of viscose rayon and cellophane. It reacts with nitrogen monoxide, NO, to form a yellow solid and equal volumes of two gaseous products.

What are the products of the reaction?

- A** S, CO, N_2
B S, CO, NO_2
C S, CO_2 , N_2
D S, CO_2 , NO_2

Answer: C

Only option C can form a balanced chemical equation with the criteria that of two gases are of equal volumes.

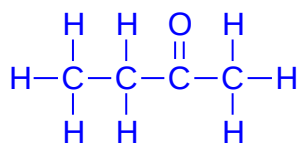
- 3 *Use of the Data Booklet is relevant to this question.*

How many hydrogen atoms are present in 4.5 g of butanone?

[L = Avogadro constant]

- A** 72L **B** L **C** $\frac{L}{2}$ **D** $\frac{L}{16}$

Answer: C



Mr = 72.0 g

$$\text{Amount of butanone} = \frac{4.5}{72} = \frac{1}{16} \text{ mol}$$

$$\text{Amount of hydrogen atoms} = \frac{1}{16} \times 8 = \frac{1}{2} \text{ mol}$$

$$\text{Number of hydrogen atoms} = \frac{L}{2}$$

- 4 A 10 cm³ gaseous sample of butanethiol, CH₃(CH₂)₃SH, was sparked in 100 cm³ of oxygen to produce carbon dioxide, sulfur dioxide and water. After cooling to room temperature, the residual gas mixture was passed through aqueous potassium hydroxide.

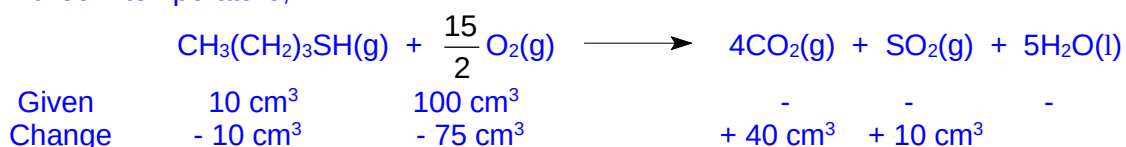
What is the final volume of the resultant gas at room temperature and pressure after passing through the alkali?

- A 25 cm³ B 35 cm³ C 50 cm³ D 75 cm³

Answer: A

Need to write the balanced equation first.

At room temperature,



Gases absorbed by alkali are CO₂(g) and SO₂(g)

Volume of residual gas mixture = volume of excess of O₂ = **25 cm³**

- 5 Paramagnetism is a relatively weak magnetic force produced by unpaired electrons. It is only detectable in the presence of external magnetic field. The strength of the force is directly proportional to the number of unpaired electrons in an atom or ion.

Which atom has the greatest paramagnetism?

- A magnesium B silicon **C** phosphorus D sulfur

Answer: C

¹²Mg: [Ne] 3s² (0 unpaired electrons)

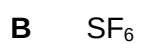
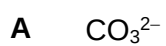
¹⁴Si: [Ne] 3s² 3p² (2 unpaired electrons)

¹⁵P: [Ne] 3s² 3d³ (3 unpaired electrons)

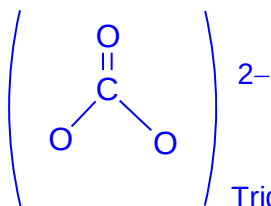
¹⁶S: [Ne] 3s² 3p⁴ (2 unpaired electrons)

Since P has the greatest number of unpaired electrons, it has the greatest paramagnetism.

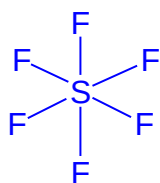
6 Which species has the biggest bond angle?



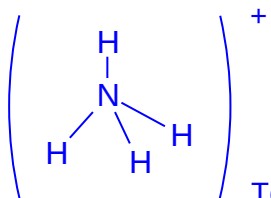
Answer: A



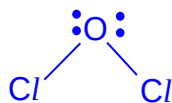
Trigonal planar shape, bond angle = 120°



Octahedral shape, bond angle = 90°



Tetrahedral shape, bond angle = 109.5°



Bent shape, bond angle = 104.5°

- 7 Which of the following shows the correct response for the presence (indicated by ✓) or absence (indicated by x) of hydrogen bonds in the substances?

	$\text{NH}_3(\text{l})$	$\text{NH}_4\text{Cl}(\text{s})$	$\text{CH}_3\text{CONH}_2(\text{l})$
A	✓	✓	✓
B	✓	x	✓
C	x	✓	x
D	✓	x	x

Answer: B

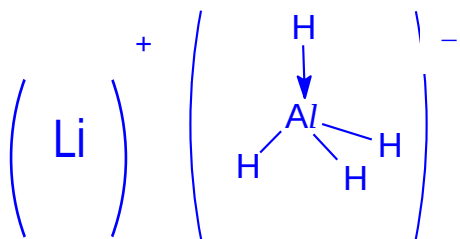
$\text{NH}_3(\text{l})$	$\text{NH}_4\text{Cl}(\text{s})$	$\text{CH}_3\text{CONH}_2(\text{l})$
Hydrogen bond between ammonia molecules	Ionic bond between the NH_4^+ and Cl^- ions	Hydrogen bond between amide molecules

- 8 Which types of bonding are present in lithium aluminum hydride, LiAlH_4 ?

- 1 co-ordinate (dative covalent)
- 2 ionic
- 3 metallic

- A 1 only B 1 and 2 only C 2 and 3 only D 1, 2 and 3

Answer: B



There are ionic bonds (electrostatic forces of attraction) between Li^+ cation and AlH_4^- anion.

Hydrogen atom being more electronegative than Al, will gain the extra electron forming a hydride, H^- ion. H^- ion shares its lone pair of electrons with Al atom by forming a dative bond with it.

No metallic bonding as this is a giant ionic structure.

- 9 A 25 cm³ solution containing of 0.20 mol dm⁻³ Ti⁺ required 25 cm³ of 0.10 mol dm⁻³ acidified KMnO₄ to oxidise it to Ti³⁺ in solution.

What is the oxidation state of the manganese in the reduced form?

- A 2 B 3 C 4 D 7

Answer: B



$$\frac{n(\text{Ti}^+)}{n(\text{MnO}_4^-)} = \frac{(0.20)(0.025)}{(0.10)(0.025)} = 2$$

$$\frac{n(\text{e}^-)}{n(\text{MnO}_4^-)} = \frac{4}{1}$$

[R] 1 mol MnO₄⁻ takes in 4 mole of e⁻. Hence, oxidation state of Mn decreases by 4. Mn(VII) in MnO₄⁻ reduced to Mn(III).

- 10 Sodium, silicon, sulfur and chlorine are elements in Period 3 of the Periodic Table. Which of the following statements about these elements are true?

- 1 Sodium oxide has a giant lattice of ions.
- 2 Oxides of sulfur dissolve in water to form a solution that turns litmus red.
- 3 The chlorides of silicon have a higher melting point than that of oxides of silicon.
- 4 Sulfur exists as a yellowish gas as it has lower melting point than chlorine.

- A 1 and 2 only
B 1, 2 and 3 only
C 2 and 4 only
D 1, 3 and 4 only

Answer: A

Statement 1: Correct. The Na⁺ and O²⁻ ions are arranged in an orderly manner in a giant ionic lattice.

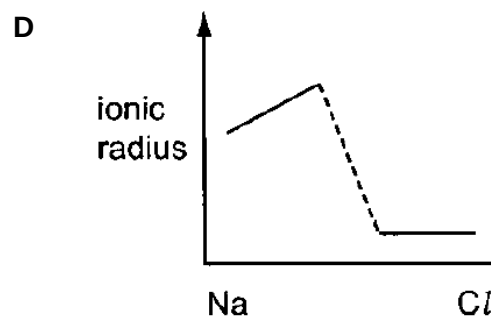
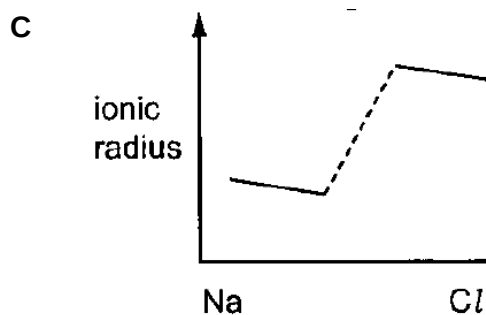
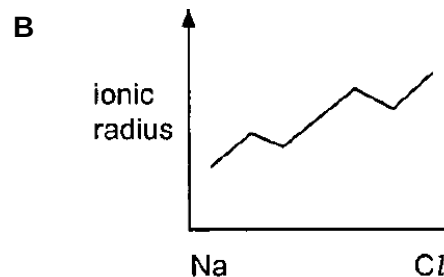
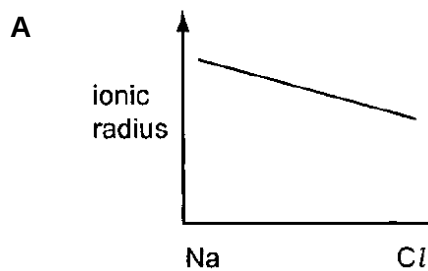
Statement 2: Correct. Oxides of sulfur are SO₂ and SO₃ which dissolved in water to form acidic H₂SO₃ and H₂SO₄ solution turning litmus red.

Statement 3: Incorrect.

Chlorides of silicon are simple molecular in nature with instantaneous dipole-induced dipole intermolecular forces. Hence they have lower melting points than **giant molecular SiO₂** structures where the S and O atoms are bonded together with **strong covalent bonds** which require a large amount of energy to overcome.

Statement 4: Incorrect. Both S₈ and Cl₂ exist as simple non-polar molecules with instantaneous dipole-induced dipole intermolecular forces. As S₈ molecules are larger with more electrons, the strength of the instantaneous dipole-induced dipole intermolecular forces are stronger. It has higher melting point than chlorine hence exist as a yellow solid.

- 11 Which diagram represents the change in ionic radius of the ions of the element across the third period (Na to Cl)?



Answer: C

Na to Al lose their valence shells to form cations while Si to Cl retained the valence shell, resulting in ionic size of anions larger than cations.

The size of cations and anions decreases across the period. This is due to increasing nuclear charge but similar shielding effect resulting to increasing strength of electrostatic forces of attraction between nucleus and electrons in the outer shell.

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- 12 The dissociation constant, K_w , for the ionisation of water, $2\text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$, at different temperatures is given below.

temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
0	1.15×10^{-15}
25	1.00×10^{-14}
50	5.50×10^{-14}

What can be deduced from this information?

- A $[\text{H}^+]$ and $[\text{OH}^-]$ are equal at 25 °C only
 B The equilibrium lies more to the left as temperature increases.
 C The $[\text{H}^+]$ increases while the $[\text{OH}^-]$ decreases as temperature increases.
 D The pH of water at 50 °C is 6.6.

Answer: D

$[\text{H}^+] = [\text{OH}^-]$ for all temperature thus (A) is wrong.

When temperature increases K_w increases. This implies position of equilibrium shifts towards right (B) favouring forward endothermic reaction (A) to absorb heat. $[\text{H}^+]$ and $[\text{OH}^-]$ increases to the same extent (C).

$$[\text{H}^+] = \sqrt{5.50 \times 10^{-14}} = 2.345 \times 10^{-7} \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg(2.345 \times 10^{-7}) = 6.6$$

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- 13 A solution was made by mixing 0.002 mol of $\text{HNO}_3(\text{aq})$ and 0.003 mol of $\text{Ba}(\text{OH})_2(\text{aq})$. Water was added until the volume of the solution was 1 dm^3 .

What is the pH of the solution?

- A 2.7 B 11.0 C 11.4 D 11.6

Answer: D

$$n(\text{OH}^-) = 0.003 \times 2 = 0.006 \text{ mol}$$

$$n(\text{OH}^-) \text{ in excess} = 0.006 - 0.002 = 0.004 \text{ mol}$$

$$\text{pH} = 14 - (-\lg 0.004) = 11.6$$

14 Which set of solutions of equal volume when mixed will give an acidic buffer?

- 1 0.05 mol dm⁻³ HCl and 0.10 mol dm⁻³ CH₃CO₂Na
- 2 0.10 mol dm⁻³ CH₃CO₂H and 0.10 mol dm⁻³ NaOH
- 3 0.10 mol dm⁻³ H₂SO₄ and 0.05 mol dm⁻³ Ca(OH)₂

- A 1 only
 B 1 and 2 only
 C 2 and 3 only
 D All

Answer: A

- 1 0.05 mol dm⁻³ HCl and 0.10 mol dm⁻³ CH₃CO₂Na
 CH₃CO₂Na is in excess.



⇒ 0.025 mol dm⁻³ CH₃CO₂H and 0.025 mol dm⁻³ CH₃CO₂Na present (buffer)

- 2 0.10 mol dm⁻³ CH₃CO₂H and 0.10 mol dm⁻³ NaOH
 Equal amount of acid and base.



⇒ only a salt of weak acid present (not buffer)

- 3 Salts of strong acids and bases do not form buffer.
 (not buffer)

15 Xenon enhanced CT scanning is an easy and non-invasive method in which patients inhale the stable (¹³¹Xe) gas while CT images are made. The CT scan is used in the study of pulmonary function and organ blood flow.

Xenon-131 is a radioactive isotope and it has a half-life of 11.8 days.

Given that radioactive decay is a first-order reaction, what fraction of the isotope has **reacted** after 59 days?

- A 0.0313 B 0.125 C 0.937 D 0.969

Answer: D

Since half-life of Xenon-131 is 11.8 days,

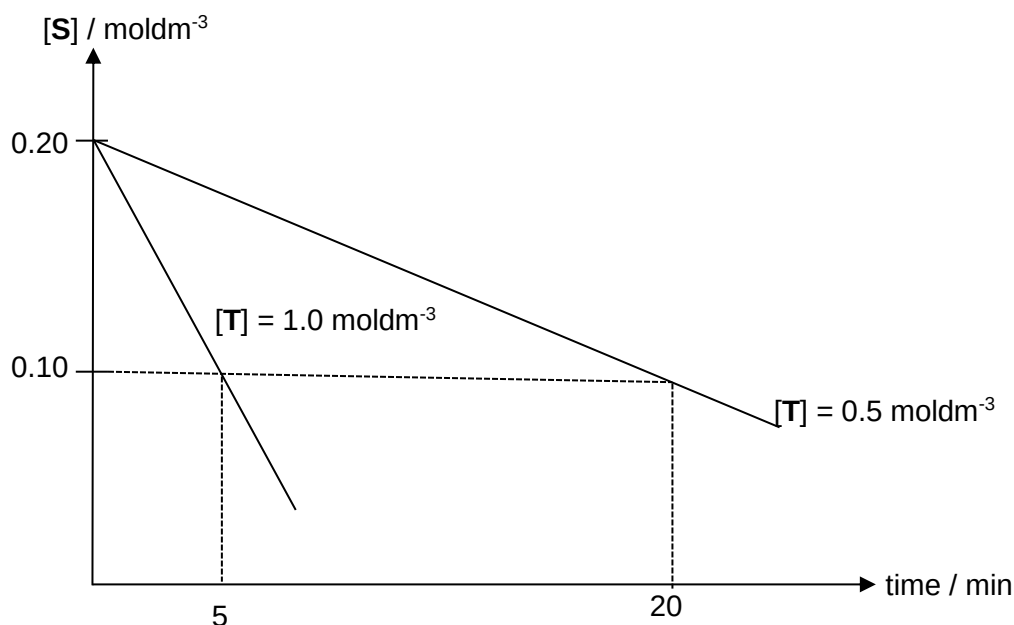
$\frac{59}{11.8} = 5$ half-lives have passed after 59 days.

Fraction **remaining** after 59 days = $\left(\frac{1}{2}\right)^5 = \frac{1}{32}$

Therefore, fraction of isotope **reacted**

$$= 1 - \frac{1}{32} = \mathbf{0.969}$$

- 16 To monitor the kinetics of a reaction between **S** and **T**, two experiments were conducted to monitor the concentration of **S** over time with different concentrations of **T**. The graph below shows the results obtained.



What is the rate equation for the reaction between **S** and **T**?

- A rate = $k [\text{S}]$
- B rate = $k [\text{T}]^2$
- C rate = $k [\text{T}][\text{S}]$
- D rate = $k [\text{T}]$

Answer: B

The gradient of each graph will give the rate of the reaction.

Given that it's a straight line, gradient is constant which mean rate is constant when concentration of **S** is changing. Hence order of reaction w.r.t. **S** is **0**

When concentration of **T** was doubled from 0.5 to 1 moldm⁻³, initial rate, which is the gradient of the graph, increase 4 times.

$$\text{Rate}_1 = \frac{0.2-0.1}{20} = 0.005 \text{ moldm}^{-3} \text{ min}^{-1}$$

$$\text{Rate}_2 = \frac{0.2-0.1}{5} = 0.02 \text{ moldm}^{-3} \text{ min}^{-1}$$

Hence order of reaction w.r.t **T** is 2.

- 17 The rate of a reaction is measured in $\text{mol dm}^{-3} \text{ min}^{-1}$.

Which row is correct?

	Overall order of reaction	Units of the rate constant
A	zero	$\text{mol}^{-1} \text{ dm}^{-3} \text{ min}^{-1}$
B	first	s^{-1}
C	second	$\text{mol}^{-1} \text{ dm}^3 \text{ min}^{-1}$
D	third	$\text{mol}^{-3} \text{ dm}^9 \text{ min}^{-1}$

Answer: C

$$\text{Rate} = k [\text{X}]^n$$

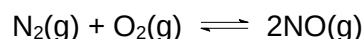
$$\text{When } n = 0, \text{ unit of } k = \text{mol dm}^{-3} \text{ min}^{-1} / (\text{mol dm}^{-3})^0 = \text{mol dm}^{-3} \text{ min}^{-1}$$

$$\text{When } n = 1, \text{ unit of } k = \text{mol dm}^{-3} \text{ min}^{-1} / (\text{mol dm}^{-3}) = \text{min}^{-1}$$

$$\text{When } n = 2, \text{ unit of } k = \text{mol dm}^{-3} \text{ min}^{-1} / (\text{mol dm}^{-3})^2 = \text{mol}^{-1} \text{ dm}^3 \text{ min}^{-1} \text{ (correct)}$$

$$\text{When } n = 3, \text{ unit of } k = \text{mol dm}^{-3} \text{ min}^{-1} / (\text{mol dm}^{-3})^3 = \text{mol}^{-2} \text{ dm}^6 \text{ min}^{-1}$$

- 18 A sample of nitrogen monoxide gas is maintained at a constant temperature of 500 K. A dynamic equilibrium is established with nitrogen, oxygen and nitrogen monoxide all present.



Which statement describes this situation?

- A At equilibrium, there are more NO gas present than N_2 and O_2 gas.
- B When pressure of reaction is reduced, more NO gas is produced.
- C Adding more NO gas after equilibrium has been established will shift position of equilibrium to the right.
- D At equilibrium, NO is formed at the same rate as it decomposes.

Answer: D

- A This statement may not be true as the amount of reactants and products formed depends on the K_c value.
- B $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$
Since there are equimolar gaseous reactants and products, the equilibria is not affected by pressure.
- C Adding more NO gas will shift equilibrium position to the left to decrease [NO].
- D At equilibrium, the concentration of both reactants and products remained constant.
This means that the rate of forward (to form NO gas) and backward (decompose to form N_2 and O_2 gas) reaction is the same.

- 19 Given that the following reversible reaction take place at T °C,



Which of the following factor affects the K_c value?

- A Decrease in temperature.
- B Addition of suitable catalyst.
- C Increase in the concentration of X.
- D Addition of water

Answer: A

K_c is only affected by temperature.

- 20 For which of the following reaction is the enthalpy change of reaction *endothermic*?

- 1 $\text{H}_2\text{SO}_4 (\text{aq}) + 2 \text{NaOH} (\text{aq}) \rightarrow \text{Na}_2\text{SO}_4 (\text{aq}) + 2 \text{H}_2\text{O} (\text{l})$
- 2 $\text{Cl}_2 (\text{g}) \rightarrow 2 \text{Cl} (\text{g})$
- 3 $\text{Mg} (\text{g}) \rightarrow \text{Mg}^+ (\text{g}) + \text{e}$
- 4 $\text{C}(\text{s}) + \text{O}_2 (\text{g}) \rightarrow \text{CO}_2 (\text{g})$

- A 1 and 4 only
- B 2 and 3 only
- C 3 and 4 only
- D 2, 3 and 4 only

Answer: B

(1) Neutralisation is exothermic.

(2) Bond breaking requires energy hence is endothermic.

(3) Removal of electron needs energy to overcome attraction forces between electrons and nucleus.

(4) Combustion of C is exothermic.

21 Which compound has the most negative lattice energy?

- A Rubidium fluoride
- B Rubidium oxide
- C Potassium fluoride
- D Potassium oxide

Answer: D

$$|\text{Lattice Energy}| \propto \left| \frac{q_+ \times q_-}{r_+ + r_-} \right|$$

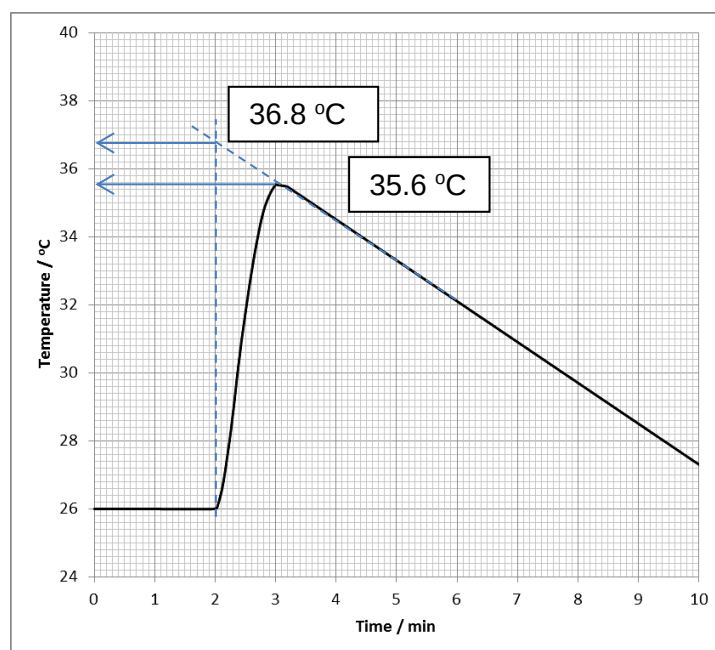
Between RbF and Rb₂O, O²⁻ has a bigger charge and hence, Rb₂O will have a more negative lattice energy.

Between KF and K₂O, O²⁻ has a bigger charge and hence, K₂O will have a more negative lattice energy.

Between Rb₂O and K₂O, K⁺ has a smaller ionic radius and hence, K₂O will have a more negative lattice energy.

- 22 A student conducted an experiment to determine the enthalpy change of solution of MgSO_4 in water. 1 mol of solid MgSO_4 was added to 100 g of water at 2 min. The changes in the temperature of the solution was monitored over a period of time.

Using the data obtained from the experiment, the following graph was obtained.



What is the enthalpy change of solution of MgSO_4 based on the data obtained from the experiment?

The specific heat capacity of water is $4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$.

- A $100 \times 4.18 \times (36.8 - 26.0)$
- B $100 \times 4.18 \times (26.0 - 36.8)$
- C $100 \times 4.18 \times (35.6 - 26.0)$
- D $100 \times 4.18 \times (26.0 - 35.6)$

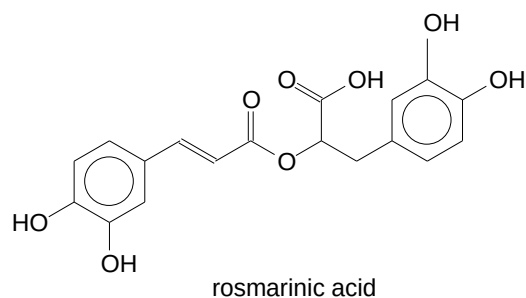
Answer: B

Since MgSO_4 was added at 2 min, the expected maximum temperature should be 36.8°C . Due to time required to heat up solution and cooling taking place, in experiment, solution will heat to only 35.6°C before cooling of solution.

FYI: The decrease in temperature after maximum temperature achieved is due to cooling taking place. This heat loss during cooling is accounted when the line is extrapolated to 2 min.

$$\Delta H = - (mc\Delta T)/n = -[100 \times 4.18 \times (36.8 - 26.0)]/1$$

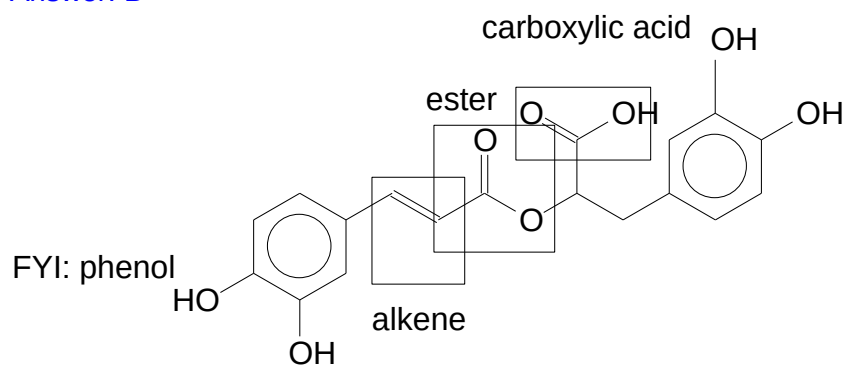
23 Rosmarinic acid occurs in culinary herbs such as rosemary, sage and thyme.



Which of the functional group is not present in rosmarinic acid?

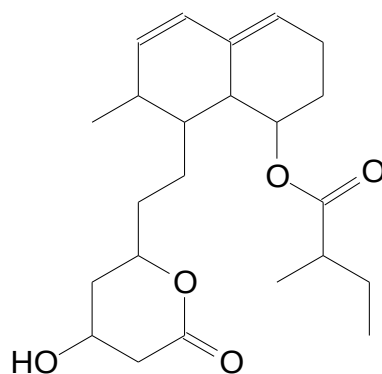
- | | |
|--------------------------|------------------|
| A ester | B alcohol |
| C carboxylic acid | D alkene |

Answer: B



Alcohol has -OH group covalently bonded to alkyl group hence can be classified as primary, secondary and tertiary alcohol. It is not present in rosmarinic acid.

- 24 Mevastatin belongs to the statin group of drugs that are used to lower cholesterol levels in humans.



What is the molecular formula of the compound?

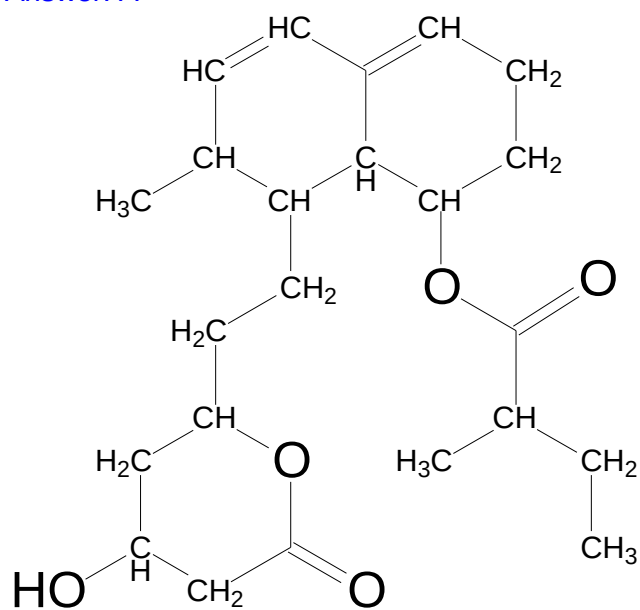
A $C_{23}H_{34}O_5$

B $C_{23}H_{31}O_5$

C $C_{22}H_{27}O_5$

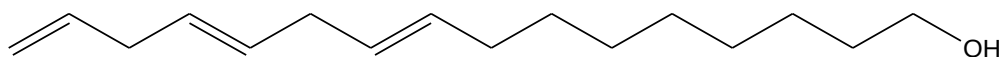
D $C_{22}H_{23}O_5$

Answer: A



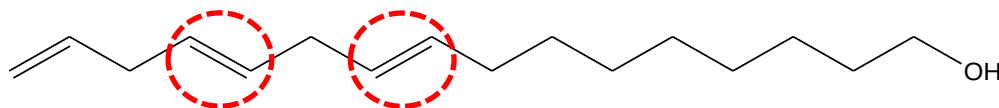
Tip: Just count the H will do 😊

- 25 How many cis-trans isomers are present in this molecule?



- A 2 B 3 C 4 D 8

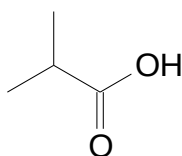
Answer: C



C=C which exhibit cis-trans isomerism are circled as shown above.

No of cis-trans isomer = $2^2 = 4$

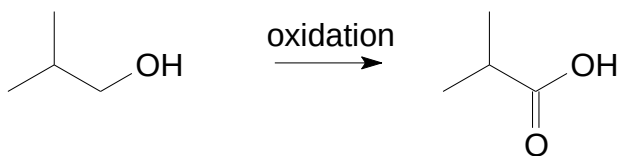
- 26 The carboxylic acid shown below can be produced by oxidation of an alcohol.



Which alcohol can be oxidised to produce this carboxylic acid when heated under reflux in the presence of an oxidising agent?

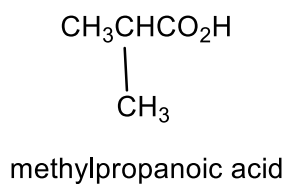
- A 2-methylpropan-2-ol
B 2-methylpropan-1-ol
C 2-methylbutan-1-ol
D butan-1-ol

Answer: B



2-methylpropan-1-ol

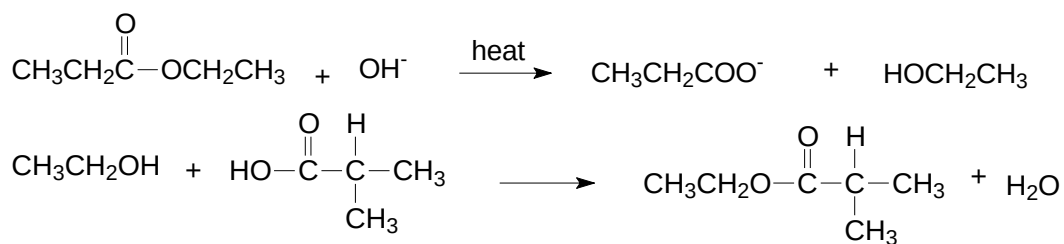
- 27 Ethyl propanoate is refluxed with aqueous sodium hydroxide. The alcohol produced is then reacted with methylpropanoic acid to make a second ester.



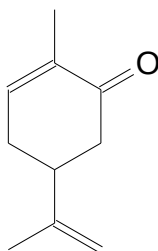
What is the structural formula of this second ester?

- A**
- $$\text{CH}_3-\text{CH}_2-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_3$$
- B**
- $$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{CH}_2-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_3$$
- C**
- $$\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_3$$
- D**
- $$\text{CH}_3-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{CH}_2-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_3$$

Answer: A



28 Carvone is responsible for the odour of spearmint.



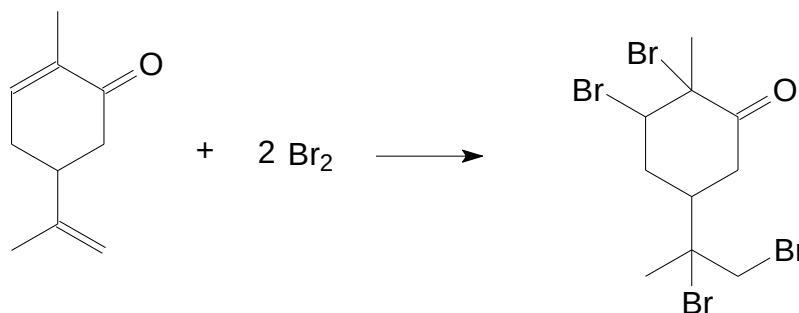
carvone

Which reagent and conditions, when added to carvone, would give observable changes?

- A Hot acidified potassium dichromate
- B Aqueous bromine
- C Warm dilute NaOH
- D Ethanoic acid with concentrated H_2SO_4 , heat

Answer: B

Only aq Br_2 reacts with alkene in carvone. Orange bromine decolourised.



29 Which of the following statements are true for thermoplastic polymers but not true for thermoset polymers?

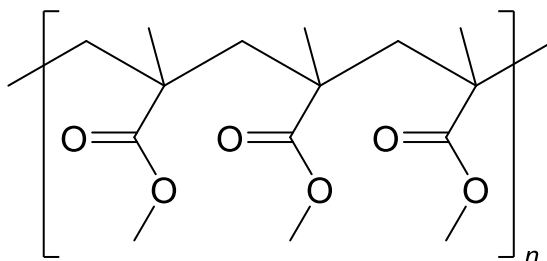
- 1 The polymer burns upon heating.
- 2 The polymer can be recycled.
- 3 There is little or no cross-linkages between polymer chains.

- A** 1 only
B 1 and 2 only
C 2 and 3 only
D 3 only

Answer: C

	Statement	Reason
1	Both true.	Both polymers can burn on heating.
2	True for thermoplastic polymer but not true for thermosetting polymer.	Thermoplastic polymers can melt and remould into different shapes to be recycled. Thermosetting polymers once the shape is set will not be able to melt and remould. Hence, it cannot be recycled.
3	True for thermoplastic polymer but not true for thermosetting polymer.	Thermoplastic polymers are formed with little or no cross-linkages hence can melt. Thermosetting polymers are formed with many cross-linkages thus the shape is set.

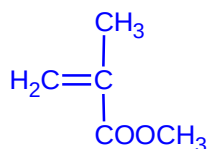
- 30 Poly(methyl methacrylate) is used to make hard contact lenses. Part of its polymer chain is shown.



Which statement about poly(methyl methacrylate) is correct?

- A A small molecule of water is lost when polymerisation took place.
- B Its monomer includes a dicarboxylic acid and a diol.
- C It can undergo alkaline hydrolysis.
- D It can form hydrogen bonds between two polymer chains.

Answer: C



The monomer $\text{CH}_2=\text{C}(\text{CH}_3)\text{COOCH}_3$ undergoes addition polymerisation to form poly(methyl methacrylate). Hence, there is no loss of small molecules of water. In the polymer, there is no hydrogen atom bonded to F/O/N atoms, hence, there is no hydrogen available for hydrogen bonding with the neighbouring chain.

The presence of an ester group in the side chain causes the polymer to hydrolyse when alkali is added.

Answer

1	A	6	A	11	C	16	B	21	D	26	B
2	C	7	B	12	D	17	C	22	B	27	A
3	C	8	B	13	D	18	D	23	B	28	B
4	A	9	B	14	A	19	A	24	A	29	C
5	C	10	A	15	D	20	B	25	C	30	C